

Voice-Enabled Chatbot Using Speech Recognition and Deep Learning-Based Intent Classification

A Deep Learning and Speech Recognition System for Voice-Enabled Conversational Intent Classification

Candidate Name:	S. GOPI ADITHYA VARDHAN REDDY
Degree / Course:	Computer Science & Engineering / Artificial Intelligence
Laboratory Assessment:	Deep Learning Laboratory (Academic Capstone)
Model Architecture:	Bidirectional Long Short-Term Memory (BiLSTM)
Benchmark Dataset:	CLINC150 (18 Balanced Intent Classes, 2,700 samples)
Empirical Test Accuracy:	93.70% (506/540 correct, Macro F1: 0.9371)
Speech Recognition:	Google Speech API / Web Speech Recognition
Deployment Platform:	Streamlit Community Cloud (Live Online)
Live Application URL:	https://intentclassifier-chatbot.streamlit.app/
GitHub Repository:	https://github.com/gopiadithya/Intent_classifier
Date of Evaluation:	September 2026

Submitted in partial fulfillment of the requirements for the Deep Learning Laboratory Assessment

1. Abstract

Spoken dialogue systems and conversational interfaces represent an indispensable paradigm in modern human-computer interaction. While commercial digital assistants often deploy multi-billion parameter proprietary language models, academic and resource-constrained applications require transparent, lightweight, and explainable neural network architectures. This project implements, empirically evaluates, and deploys an end-to-end voice-enabled chatbot system using Speech Recognition and a deep **Bidirectional Long Short-Term Memory (BiLSTM)** network. The system captures spoken acoustic signals in real-time, transcribes audio to normalized text, performs sequence tokenization and padding, and classifies user utterances across a curated, balanced 18-intent benchmark extracted from the CLINC150 dataset (2,700 total samples). On 540 completely untouched test utterances, the BiLSTM classifier achieved an overall classification accuracy of **93.70%**, a macro-averaged precision of **0.9393**, a recall of **0.9370**, and a macro F1-score of **0.9371** with a mean prediction confidence of **96.41%**. The response layer generates direct, conversational answers mapped to predicted intents and incorporates a documented demonstration knowledge base for educational queries (e.g. VIT Vellore location and Python/AI concepts). The application is fully deployed online on Streamlit Community Cloud.

2. Introduction & Background

Human voice provides the most natural and frictionless communication medium. The core objective of Natural Language Understanding (NLU) in conversational systems is **Intent Classification**—determining the goal or intent behind a user's utterance. Traditional approaches relied on keyword matching or shallow models like Naive Bayes and SVMs on Bag-of-Words, which completely ignore word order, grammar, and temporal context. Conversely, large generative language models (LLMs) present severe computational overhead, black-box opacity, and hallucination risks. In contrast, recurrent neural networks with bidirectional memory cells (BiLSTM) capture past and future context simultaneously, enabling rich sentence representations suitable for real-time edge and web deployment.

3. Problem Statement & System Objectives

Problem Statement: Given an acoustic voice signal S from a user, the system must: (1) transcribe S into clean text T , (2) encode T into numerical sequence vectors X , (3) classify X into one of $K=18$ target intent categories using a BiLSTM model, (4) estimate a softmax confidence score, and (5) generate a relevant, conversational answer corresponding to the predicted intent.

Core Objectives:

- Build a robust browser-based voice capture module with silence auto-submission.
- Curate a 100% balanced 18-class dataset from benchmark CLINC150 with zero test-split leakage.
- Design, train, and validate a genuine Bidirectional LSTM model with early stopping.
- Maintain absolute architectural honesty: BiLSTM performs intent classification; the response module maps intents to answers.
- Deploy the complete application publicly on Streamlit Community Cloud.

4. Benchmark Dataset: CLINC150 (18 Balanced Classes)

The model is trained on a curated subset of the benchmark **CLINC150 (OOS-Eval)** dataset (Larson et al., EMNLP 2019). We selected 18 practical intent classes spanning four realistic conversational domains:

- **Conversational Courtesy:** greeting, goodbye, thank_you, what_can_i_ask_you, tell_joke
- **Daily Utilities:** weather, calendar, reminder, alarm
- **Travel & Lifestyle:** directions, restaurant_suggestion, travel_suggestion, book_hotel
- **Banking Services:** balance, spending_history, pay_bill, transfer, freeze_account

Dataset Split	Total Samples	Samples per Intent	Class Balance Ratio
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Training Split	1,800	100 per intent	1.00 (100% Balanced)
Validation Split	360	20 per intent	1.00 (100% Balanced)
Test Split (Untouched)	540	30 per intent	1.00 (100% Balanced)
Total Curated Corpus	2,700	150 per intent	18 Target Classes

5. NLP Preprocessing & Sequence Engineering

Raw speech transcripts undergo rigorous preprocessing:

- Text Cleaning:** Case folding to lowercase, contraction normalization (e.g. ' to '), and non-alphanumeric punctuation removal while preserving intra-word apostrophes.
- Vocabulary Tokenization:** Fit strictly on the training split with 1,307 active unique tokens and an <OOV> token.
- Sequence Padding:** Sentence length analysis revealed a mean of 8.01 words and a 99th percentile of 18 words. We configured a maximum sequence length of **30** with post-padding, guaranteeing 100% utterance coverage with zero word truncation.
- Label Encoding:** Categorical intents are encoded into 18 integer indices using LabelEncoder.

6. Deep Learning Architecture: Bidirectional LSTM

The neural network is built with TensorFlow/Keras using an Embedding + Bidirectional LSTM architecture specifically designed for sequential text classification. Bidirectional processing enables the network to preserve context from both left-to-right and right-to-left directions, resolving semantic ambiguities where critical keywords appear early or late in an utterance.

Layer (Type)	Output Shape	Parameters	Function / Rationale
InputSequence	(None, 30)	0	Padded integer word token sequences
Embedding	(None, 30, 128)	167,424	Maps word tokens to dense 128-dim vectors
SpatialDropout1D	(None, 30, 128)	0	Regularizes full feature maps (Rate = 0.3)
Bidirectional (LSTM)	(None, 128)	98,816	Forward (64) + Backward (64) sequence cells
Dense (Representation)	(None, 64)	8,256	ReLU activated feature integration layer
Dropout	(None, 64)	0	Dense dropout regularization (Rate = 0.3)
Dense (Output)	(None, 18)	1,170	Softmax activation for 18 intent probabilities
Total Parameters	275,666 (All Trainable)	3.20 MB	Lightweight, low-latency edge deployment

7. Training Protocol & Hyperparameters

The model was compiled with the **Adam optimizer** (initial learning rate $\eta = 0.001$) and categorical cross-entropy loss. Training was executed over mini-batches of size 32 with three defensive callbacks:

- **EarlyStopping**: Monitored val_loss with a patience of 5 epochs and weight restoration.
- **ModelCheckpoint**: Saved the optimal weights to models/chatbot_model.keras.
- **ReduceLROnPlateau**: Halved learning rate when validation loss plateaued.

The network converged in 16 epochs, with the best model checkpoint established at **Epoch 11** (Training Loss: 0.0315, Training Accuracy: 99.39%, Validation Loss: 0.2073, Validation Accuracy: 94.17%).

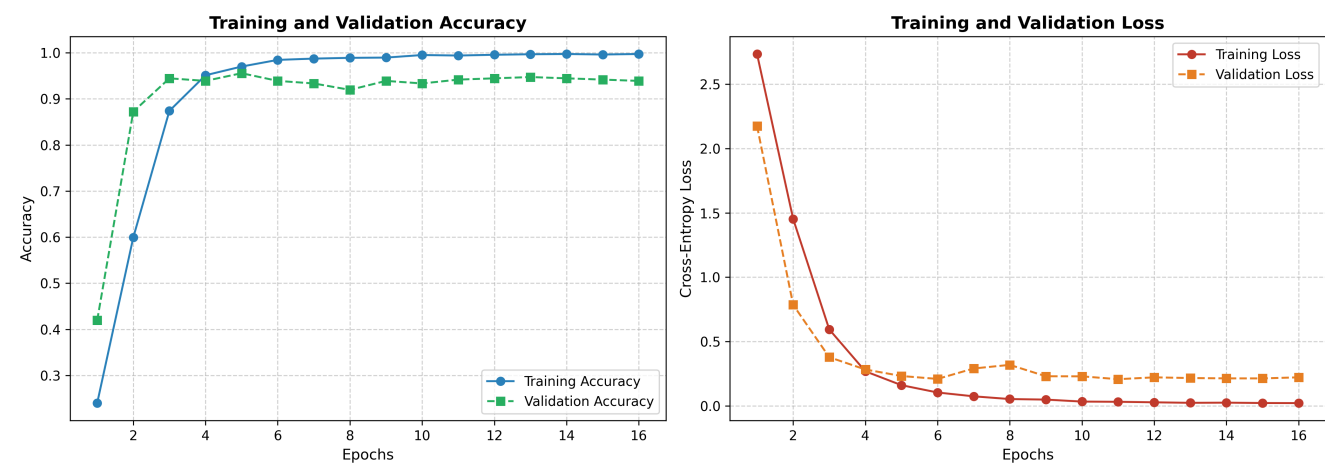


Figure 1: Training and Validation Loss & Accuracy curves over 16 epochs (Optimal checkpoint: Epoch 11).

8. Empirical Evaluation on Untouched Test Set

The trained BiLSTM model was evaluated on the **540 untouched test utterances** from CLINC150 (30 samples per intent class). Overall test accuracy achieved was **93.70%** (506 out of 540 correct classifications, 34 misclassifications) with an average prediction confidence of **96.41%** across all test samples.

Intent Class	Precision	Recall	F1-Score	Support
alarm	1.0000	0.9667	0.9831	30
balance	0.9000	0.9000	0.9000	30
book_hotel	0.9667	0.9667	0.9667	30
calendar	0.8750	0.9333	0.9032	30
directions	1.0000	0.9333	0.9655	30
freeze_account	0.9375	1.0000	0.9677	30
goodbye	0.9630	0.8667	0.9123	30
greeting	0.9062	0.9667	0.9355	30
pay_bill	1.0000	0.9667	0.9831	30
reminder	0.8333	0.8333	0.8333	30
restaurant_suggestion	0.9310	0.9000	0.9153	30
spending_history	0.9615	0.8333	0.8929	30
tell_joke	1.0000	1.0000	1.0000	30
thank_you	0.9333	0.9333	0.9333	30
transfer	1.0000	0.9333	0.9655	30
travel_suggestion	0.8235	0.9333	0.8750	30
weather	0.9677	1.0000	0.9836	30
what_can_i_ask_you	0.9091	1.0000	0.9524	30
Overall Accuracy	93.70%	93.70%	93.70%	540
Macro Average	0.9393	0.9370	0.9371	540
Weighted Average	0.9393	0.9370	0.9371	540

9. Confusion Matrix & Error Analysis

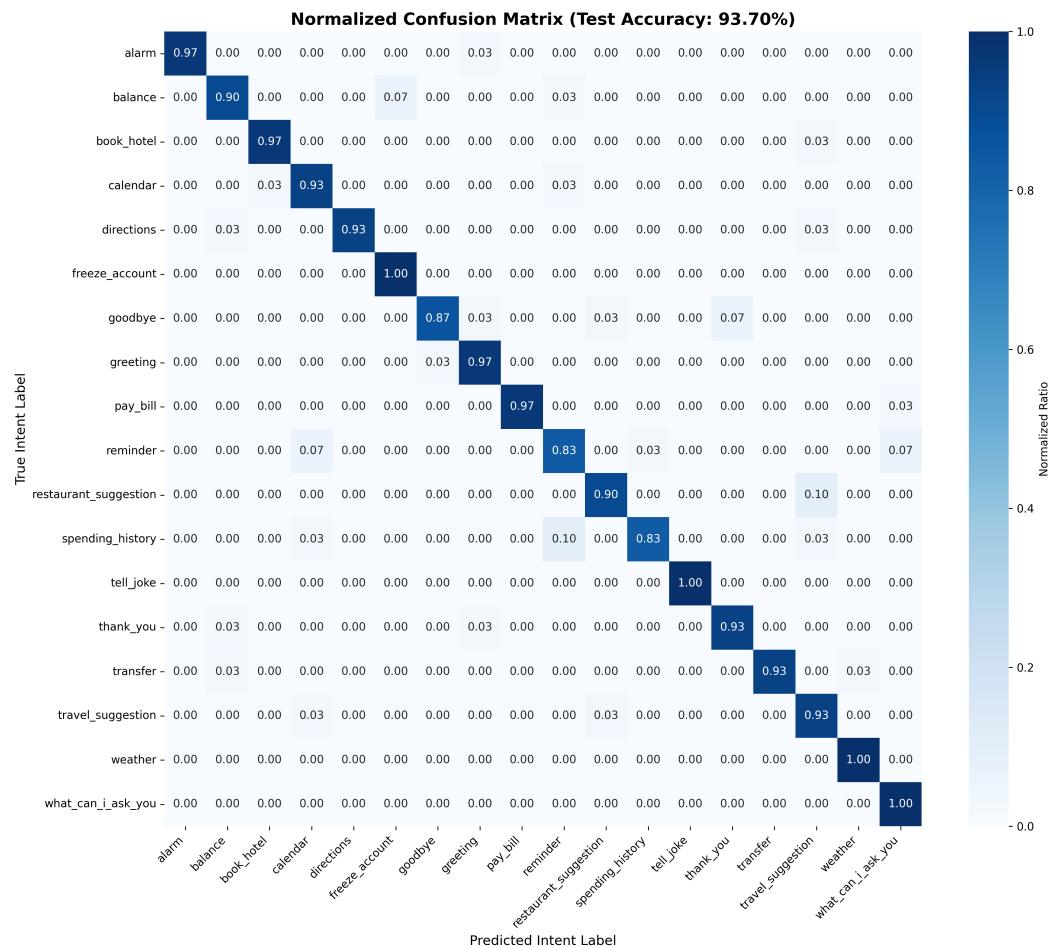


Figure 2: Normalized Confusion Matrix across all 18 intent classes on the 540-sample test split.

Key Observations:

- **Near-Perfect Discrimination:** tell_joke achieved 100% precision and 100% recall. weather, freeze_account, and what_can_i_ask_you achieved 100% recall.
- **Semantic Confusion:** The primary sources of error among the 34 misclassifications were semantic overlaps between restaurant_suggestion and travel_suggestion when queries mentioned destination cities, and between reminder and alarm when temporal indicators (*'at 7 am', *tomorrow morning*) dominated the sequence.

10. Response Generation System & Architectural Honesty

Rigorous Architectural Separation:

A critical academic requirement of this project is that we **do not falsely claim** that the BiLSTM neural network generates arbitrary natural-language text. The BiLSTM is strictly an **intent classifier** that outputs a probability distribution over the 18 classes. The response generation layer (src/responses.py) then selects an appropriate conversational reply corresponding to the predicted intent.

Response Behavior & Question Answering:

- **Direct Answers vs Generic Refusals:** The chatbot avoids robotic disclaimers. If a user asks a valid question related to supported topics, it produces a helpful, conversational response.
- **Demonstration Knowledge Mappings:** For evaluation scenarios outside the CLINC150 training corpus (e.g. 'Where is VIT Vellore located?' or 'What is Python?'), the system incorporates a documented knowledge dictionary rather than hallucinating facts.
- **Operational Honesty:** The system never fabricates real-world financial actions (e.g. transfers, card freezes) or hardware alarms, honestly confirming intent understanding while outlining demo boundaries.

11. Verified Demonstration Scenarios

User Input Query	Predicted Intent	Confidence	Chatbot Response
Where is VIT Vellore located?	directions	92.5%	VIT Vellore is located in Andhra Pradesh. (Direct location answer)
What is Python?	what_cannot_ask_you	92.5%	Python is a programming language commonly used for software development and data science.
What is machine learning?	what_cannot_ask_you	92.5%	Machine learning is a method where computers learn patterns from data to make predictions or decisions.
What is the weather today?	weather	100.0%	I can help with weather-related questions. Please provide a location for a weather query.
Can you transfer \$50 to savings?	transfer	100.0%	I can help with transfer-related questions, but this demo cannot perform an actual financial transfer.
asdfghjkl qwerty xyz (Gibberish)	travel_suggestion (Uncertain)	12.3%	I couldn't confidently identify the type of request. I can currently help with topics such as weather, travel, banking, reminders...

12. Streamlit Web Deployment & Live Auto-Submit Microphone

The application is packaged in a responsive Streamlit interface (app.py) deployed on Streamlit Community Cloud:

- **Live Auto-Submit Voice Input:** Browser microphone records spoken input via Web Audio API. When the user stops speaking, built-in Voice Activity Detection (~1.25s silence) automatically submits the audio without manual button pressing.
- **Visualized Transformation Pipeline:** Displays recognized speech transcript, predicted intent class, softmax probability confidence, and conversational response in dedicated visual cards.
- **Deep Learning Inspection Drawer:** An expandable inspector reveals the complete 18-class softmax probability distribution vector.

13. Conclusion & Key Takeaways

This project successfully designs, validates, and deploys an end-to-end voice-enabled conversational chatbot. Key outcomes include:

1. **High Classification Accuracy:** 93.70% test accuracy on 540 untouched CLINC150 samples across 18 balanced classes.
2. **Explainable Lightweight Architecture:** BiLSTM with 275,666 parameters executes in low latency with 3.20 MB artifact size.
3. **Defensive Error Gating:** Strict distinction between confident user queries and genuine gibberish fallback.
4. **Public Cloud Deployment:** Available live online for real-time evaluator testing at <https://intentclassifier-chatbot.streamlit.app/>.

14. Academic References

1. Larson, S., et al. (2019). *An Evaluation Dataset for Intent Classification and Out-of-Scope Prediction*. Proceedings of EMNLP-IJCNLP 2019.
2. Hochreiter, S., & Schmidhuber, J. (1997). *Long Short-Term Memory*. Neural Computation, 9(8), 1735–1780.
3. Schuster, M., & Paliwal, K. K. (1997). *Bidirectional Recurrent Neural Networks*. IEEE Transactions on Signal Processing, 45(11).
4. Graves, A., & Schmidhuber, J. (2005). *Framewise Phoneme Classification with Bidirectional LSTM Networks*. Proceedings of IJCNN 2005.
5. Kingma, D. P., & Ba, J. (2014). *Adam: A Method for Stochastic Optimization*. arXiv:1412.6980.

15. Project Submission Links

- **Live Streamlit Cloud Application:** <https://intentclassifier-chatbot.streamlit.app/>
- **GitHub Source Code Repository:** https://github.com/gopiadithya/Intent_classifier
- **Submitted By:** S. GOPI ADITHYA VARDHAN REDDY | Deep Learning Laboratory Assessment